Int by Parts Review: Backwards product rule!

- Integrate and rearrange the product rule to get

$$d(uv) = u\,dv + v\,du \Longrightarrow \boxed{\int u\,dv = uv - \int v\,du} + C$$

  Differentiate u, integrate dv include $+C$ in the final integral.
- Make $\int v\,du$ simpler. Useful choices that simplify when differentiated are $u = t$, $u = \ln(t)$, $u = \arctan(t)$, ...
  - Identify u and dv.
  - Differentiate u to get du. Integrate dv to get v.
  - Substitute into $\int u\,dv = uv - \int v\,du$.

Worked example $\int \ln(t)\,dt$

For $t > 0$, write the integrand as $\ln(t) \cdot 1$.

$$\boxed{\begin{aligned} u &= \ln(t) \\ du &= \frac{1}{t}\,dt \end{aligned}} \qquad \boxed{\begin{aligned} dv &= dt \\ v &= t \end{aligned}}$$

$$\int \ln(t)\,dt = t\ln(t) - \int t\,\frac{1}{t}\,dt = t\ln(t) - \int 1\,dt$$

$$\boxed{\int \ln(t)\,dt = t\ln(t) - t + C}$$

Worked example $\int te^{2t}\,dt$

$$\boxed{\begin{aligned} u &= t \\ du &= dt \end{aligned}} \qquad \boxed{\begin{aligned} dv &= e^{2t}\,dt \\ v &= \frac{e^{2t}}{2} \end{aligned}}$$

$$\int te^{2t}\,dt = \frac{te^{2t}}{2} - \int \frac{e^{2t}}{2}\,dt = \frac{te^{2t}}{2} - \frac{1}{2}\int e^{2t}\,dt$$

$$\boxed{\int te^{2t}\,dt = \frac{te^{2t}}{2} - \frac{e^{2t}}{4} + C}$$

Remember the $1/2$ each time you integrate e^{2t}.